Circuitos lógicos combinacionais

Combinação de portas lógicas

O nível lógico da saída depende da entrada

Não possui memória

Usa tabela-verdade

Somador, comparador, decodificador, multiplexador, etc.

Circuitos aritméticos

Soma, subtração, multiplicação, e divisão

Ex.: ULA de um microprocessador

Transporte de Saída (*carry*)

A	B	S	T_S
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Meio somador, *half adder* (HA)

Circuito aritmético combinacional que adiciona 2 “bits” (entradas)

Saída: 1 bit de soma e 1 bit de carry

“half” ← não considera o carry na <u>entrada</u>

Usando só “half” não é possível usar o carry no próximo somador

XOR

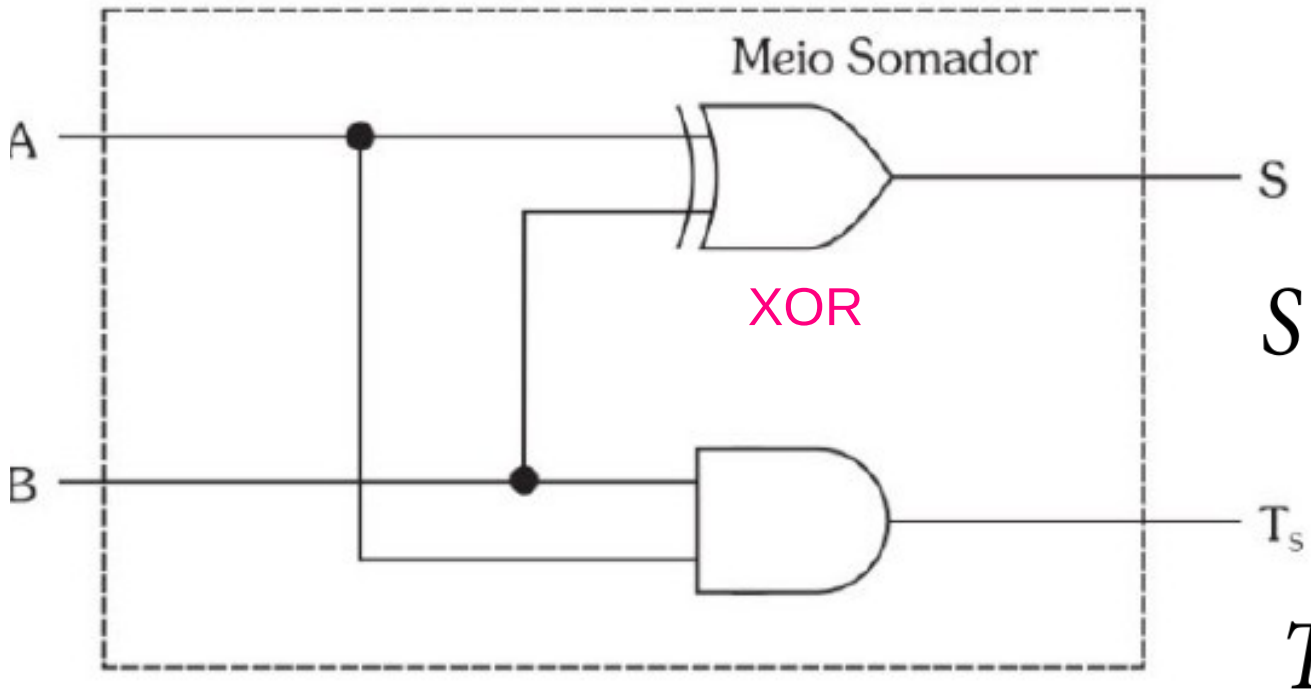
$$S = \overline{A}B + A\overline{B} = A \oplus B$$

$$T_S = AB$$

= C (carry)

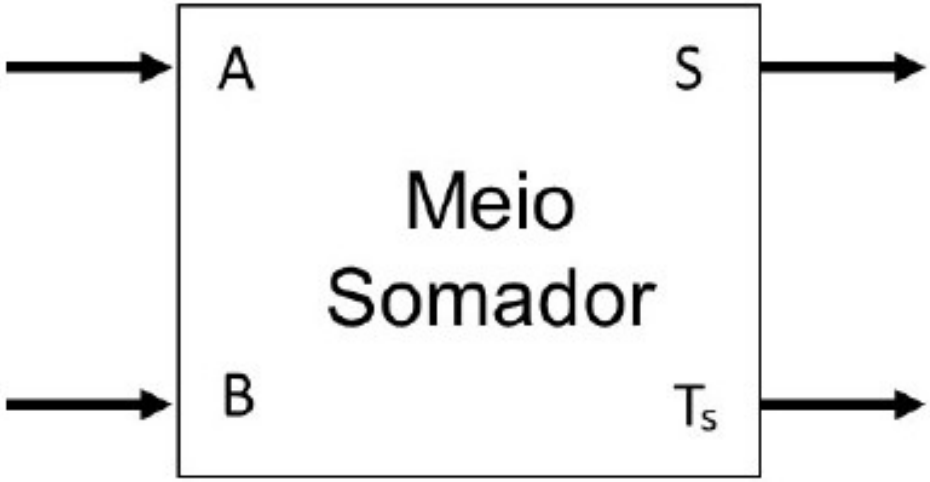
Transporte de Saída (carry, “vai um”)

Meio somador, *half adder* (HA)



$$S = \overline{A}B + A\overline{B} = A \oplus B$$

$$T_S = AB$$



Full Adder (FA)

		carry in	soma	carry out
A	B	T_E	S	T_S
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

FA - expressão característica

$$S = \overline{A}\,\overline{B}T_E + \overline{A}B\overline{T_E} + A\overline{B}\,\overline{T_E} + ABT_E$$

Obtido a partir da tabela anterior

$$S = \overline{A}\left(\overline{B}T_E + B\overline{T_E}\right) + A\left(\overline{B}\,\overline{T_E} + BT_E\right)$$

$$S = \overline{A}\left(B \oplus T_E\right) + A\left(B \odot T_E\right)$$

$$S = \overline{A}\left(B \oplus T_E\right) + A\left(\overline{B \oplus T_E}\right)$$

Simplificação

$$X = B \oplus T_E$$

$$S = \overline{A}\left(X\right) + A\left(\overline{X}\right)$$

$$S = A \oplus X$$

XOR

carry ou “vai-um” de entrada

$$\boxed{S = A \oplus B \oplus T_E}$$

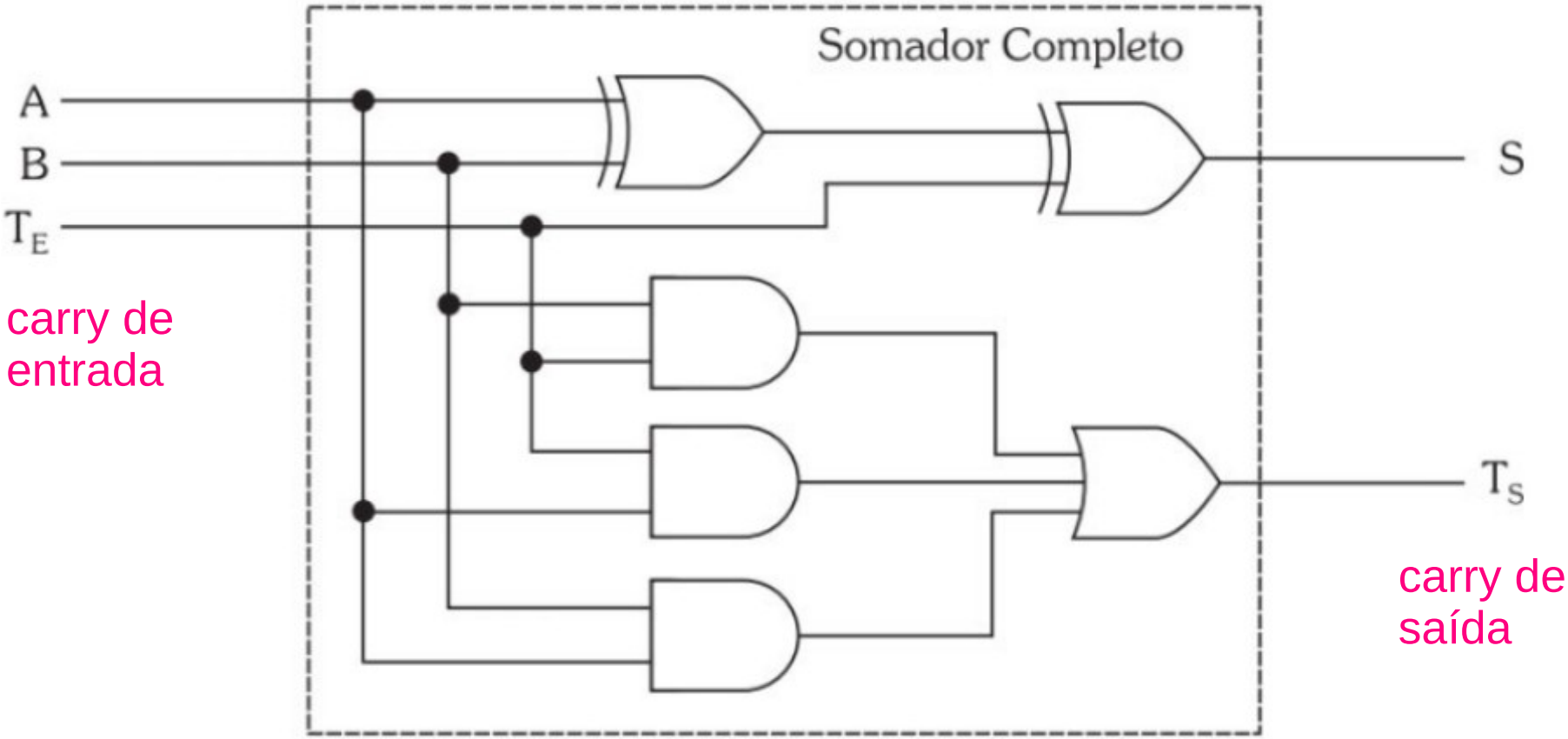
carry de saída

$$T_S = \overline{A}BT_E + A\overline{B}T_E + AB\overline{T_E} + ABT_E$$

	$\overline{B}$		B	
$\overline{A}$	0	0	1	0
A	0	1	1	1
	$\overline{T_E}$		T_E	$\overline{T_E}$

$$\boxed{T_S = AT_E + BT_E + AB}$$

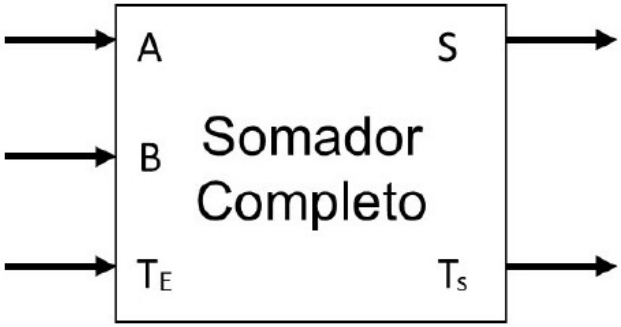
Circuito



$$S = A \oplus B \oplus T_E$$

$$T_S = AT_E + BT_E + AB$$

(dos slides anteriores)

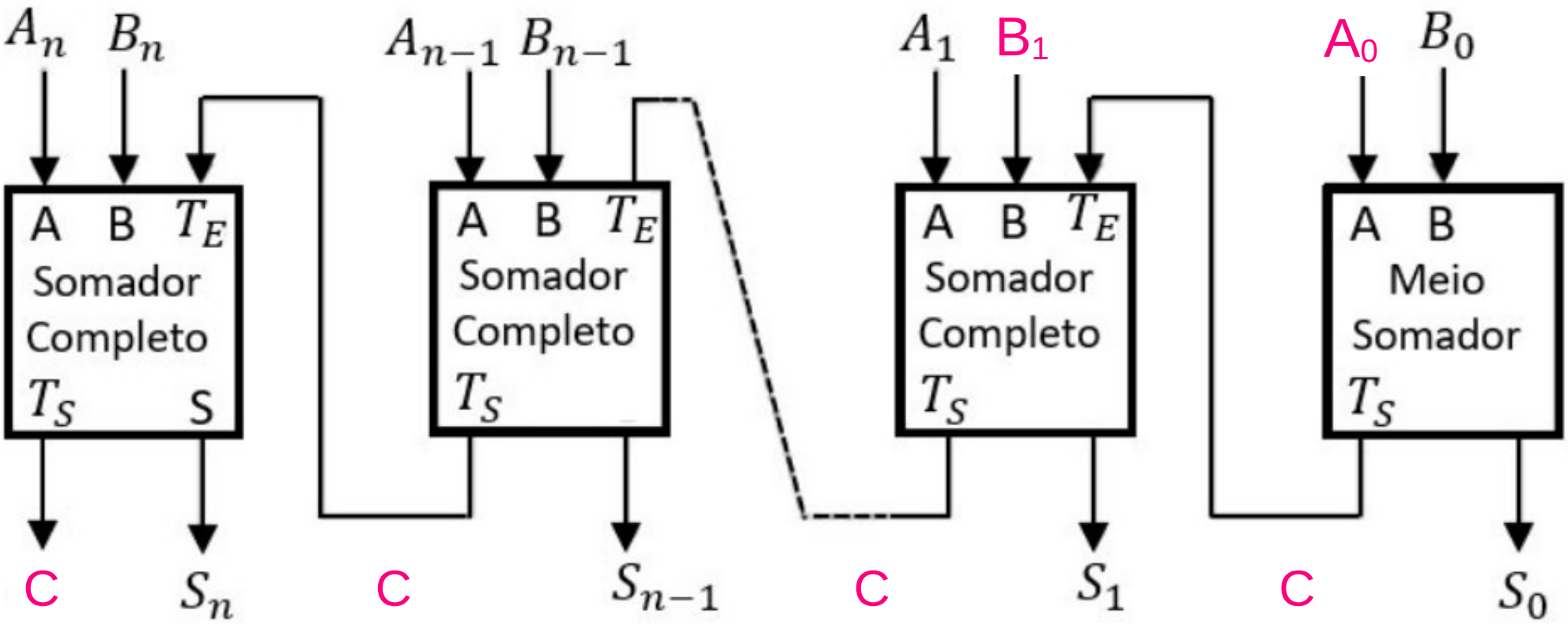


Bloco representativo

Soma de 2 nºs com *n* bits

Entrada
2 números, A e B

menos
significativo



Saída

Meio Subtrator - Half Subtractor (HS)

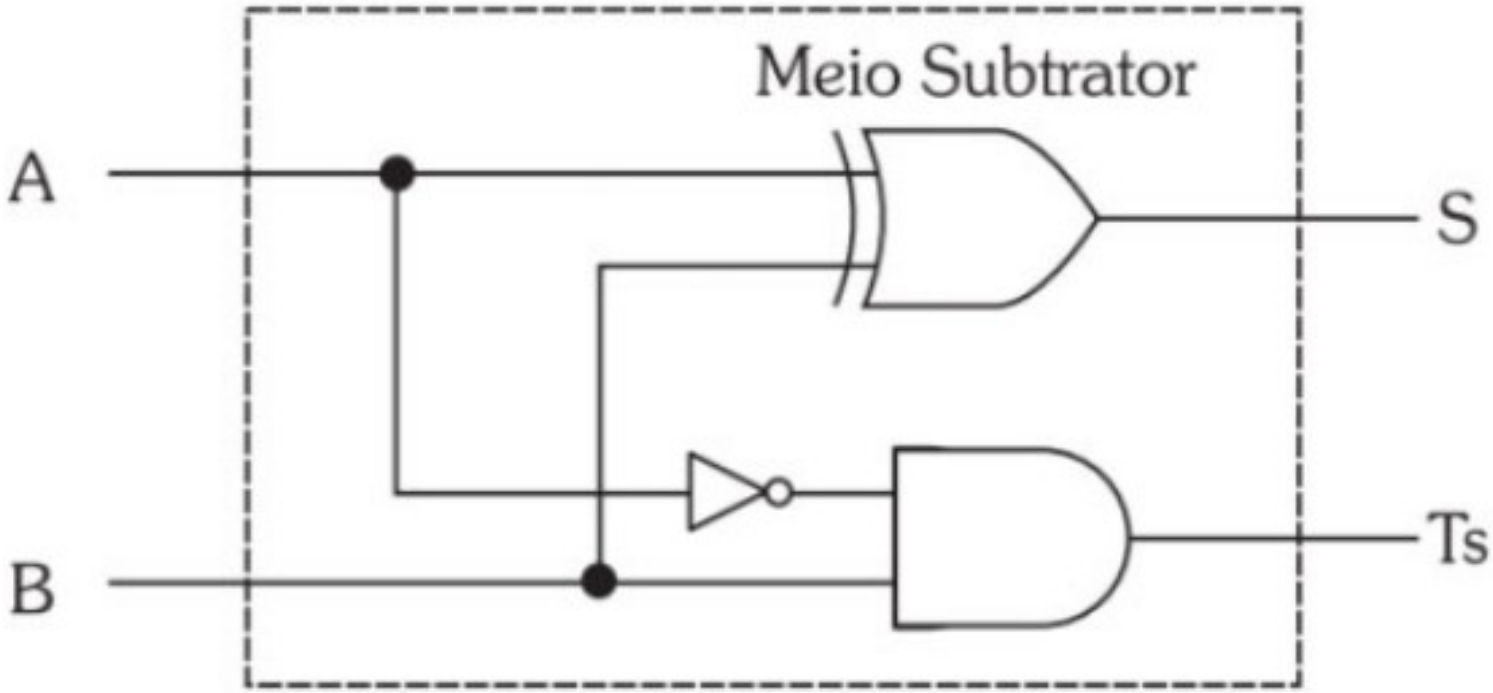
Entrada Saída

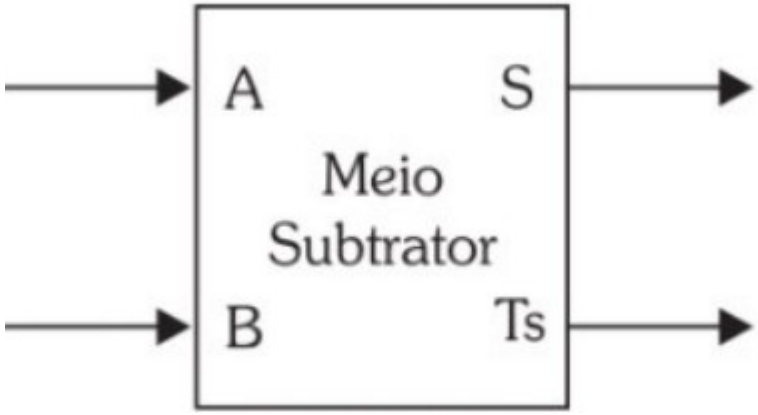
A	B	S	T_S
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$S = \overline{A}B + A\overline{B} = A \oplus B$$

$$T_S = \overline{A}B$$

Meio Subtrator





Subtrator Completo - Full Subtractor (FS)

A	B	T_E	S	T_S
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

FS - Carry

$$S = \overline{A}\,\overline{B}T_E + \overline{A}B\overline{T_E} + A\overline{B}\overline{T_E} + ABT_E$$

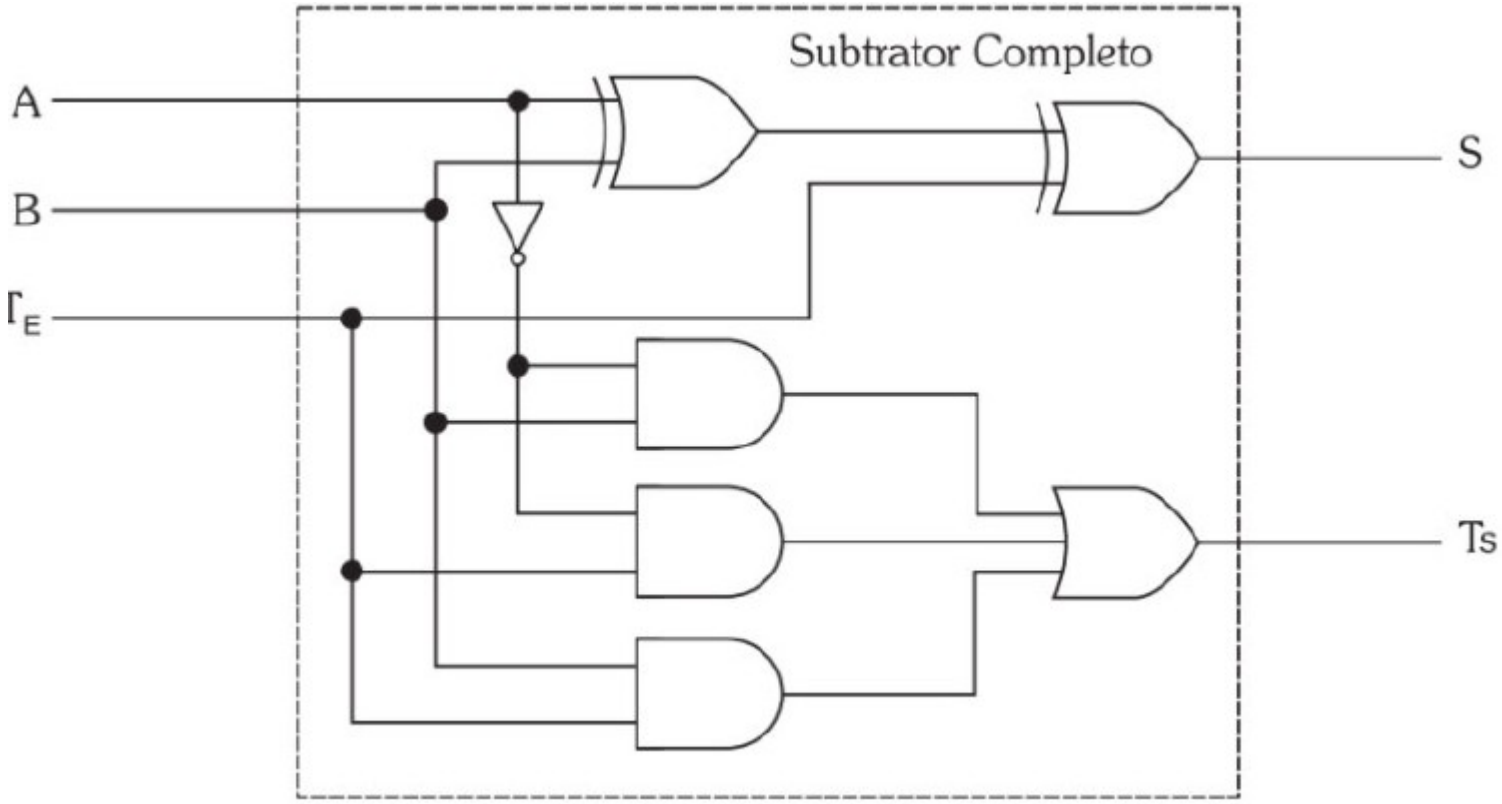
$$S = A \oplus B \oplus T_E$$

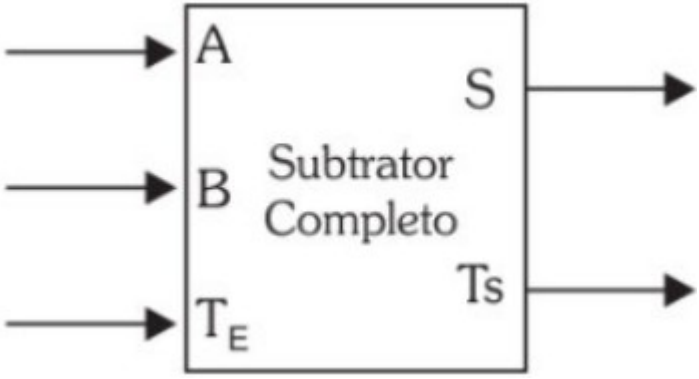
$$T_S = \overline{A}\,\overline{B}T_E + \overline{A}B\overline{T_E} + \overline{A}BT_E + ABT_E$$

	$\overline{B}$		B	
$\overline{A}$	0	1	1	1
A	0	0	1	0
	$\overline{T_E}$		T_E	$\overline{T_E}$

$$T_S = \overline{A}T_E + BT_E + \overline{A}B$$

FS

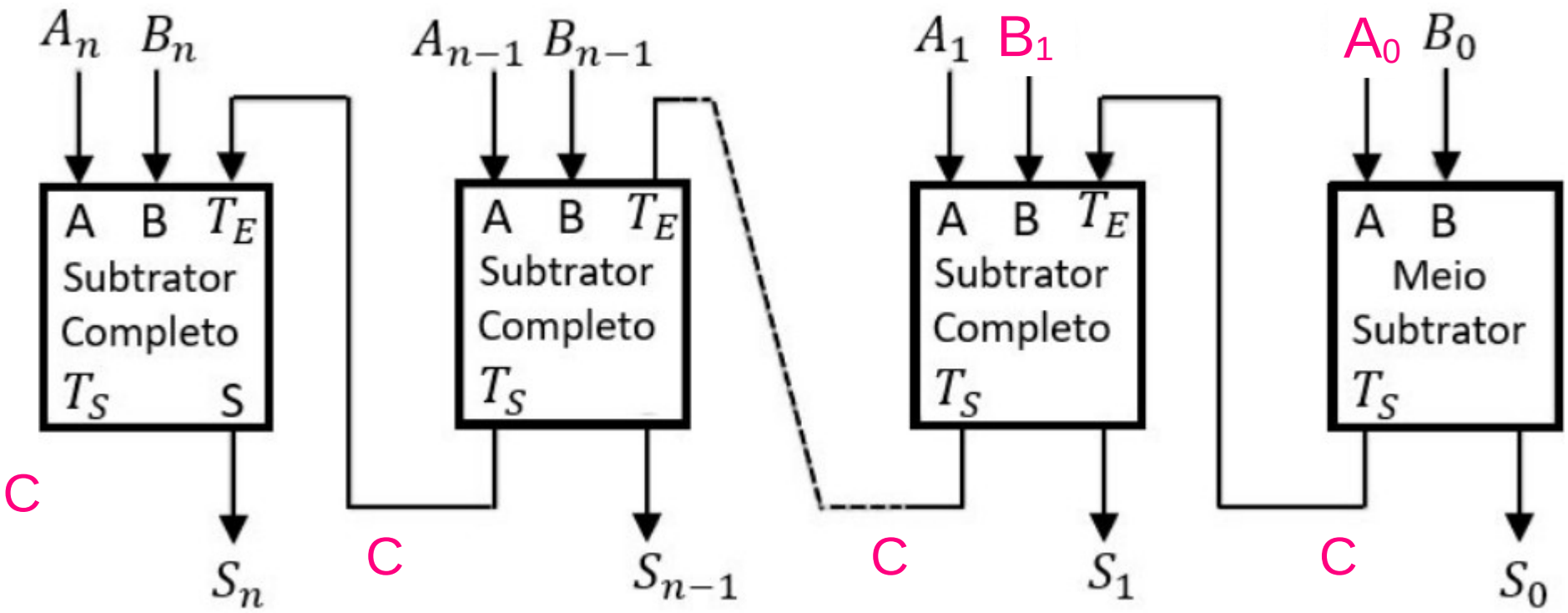




Subtrator Completo - FS

Entrada
2 números, A e B

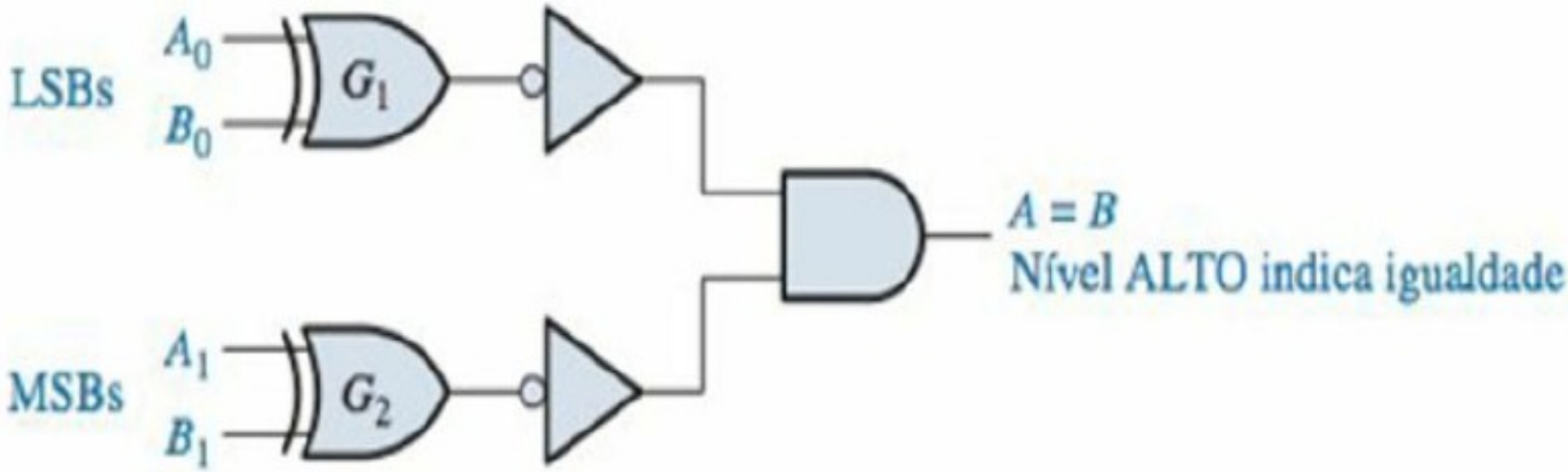
menos
significativo



Saída

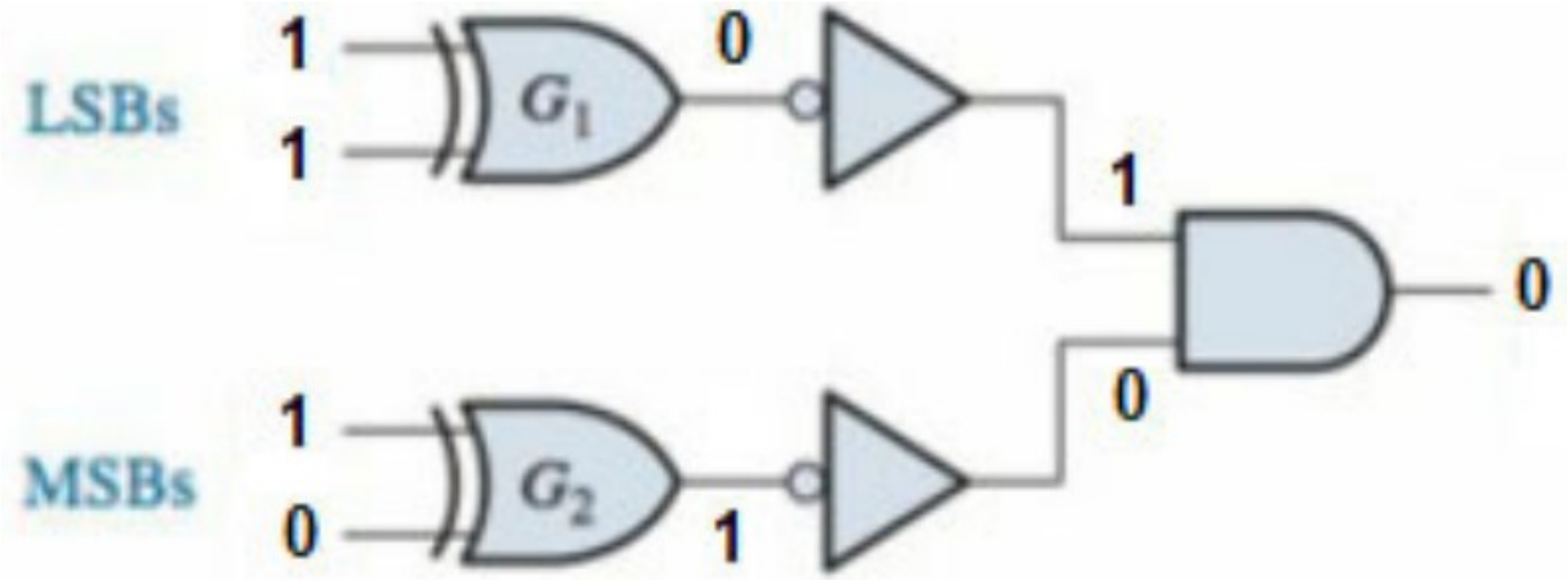
Comparador

Compara 2 n^{os} binários para determinar se são iguais



Formato geral: Número binário $A \rightarrow A_1A_0$
Número binário $B \rightarrow B_1B_0$

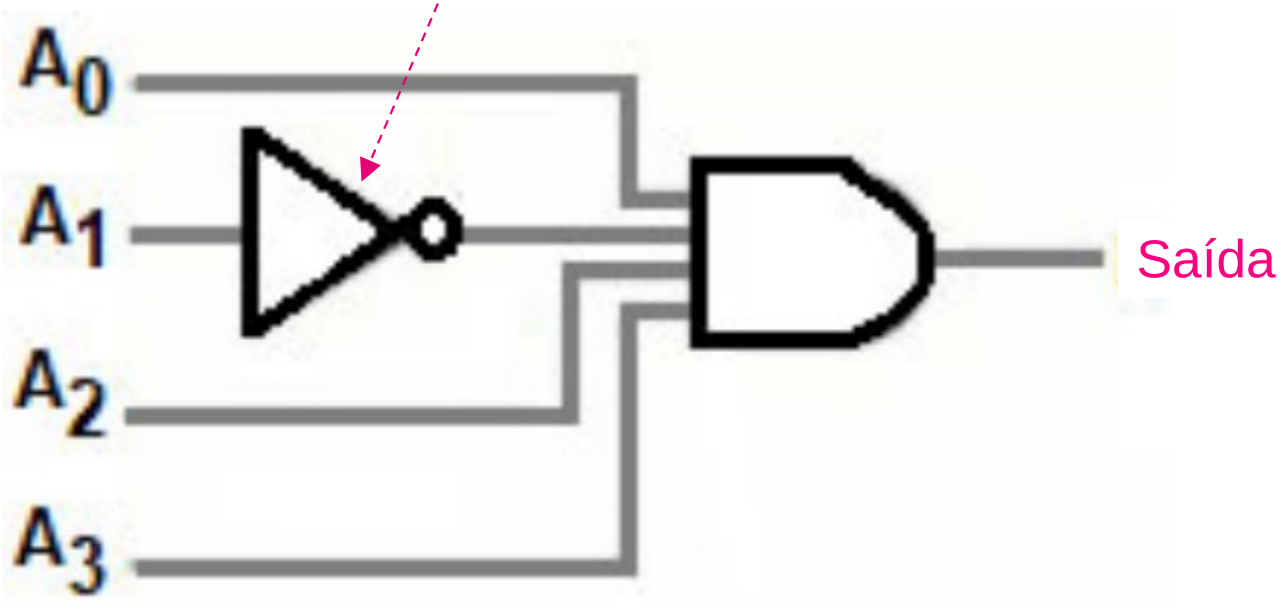
Exemplo - comparador



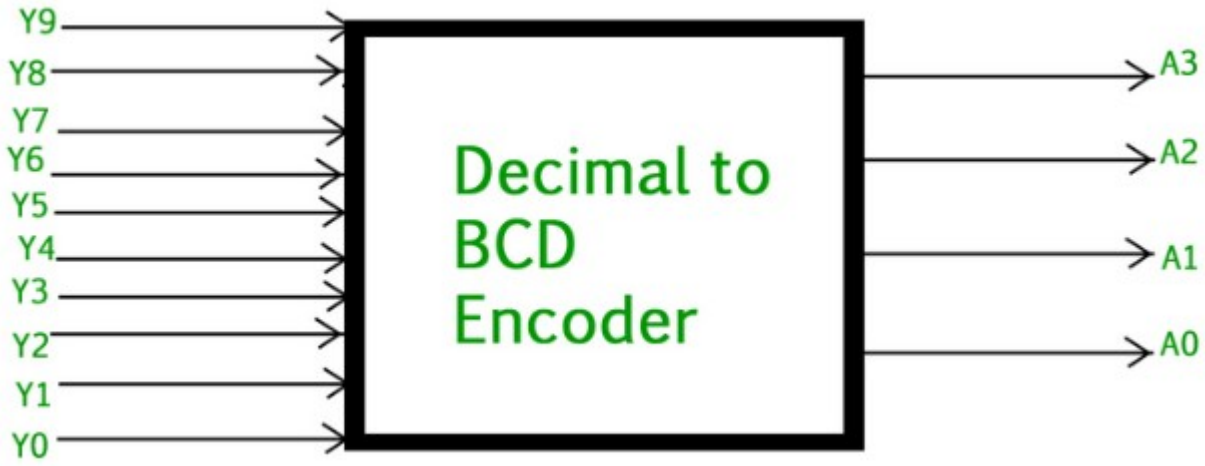
(De)codificador

Decodificar o nº 1101_{10} produzindo nível 1 na saída

$$D = A_3 A_2 \overline{A_1} A_0$$



Codificador decimal para BCD



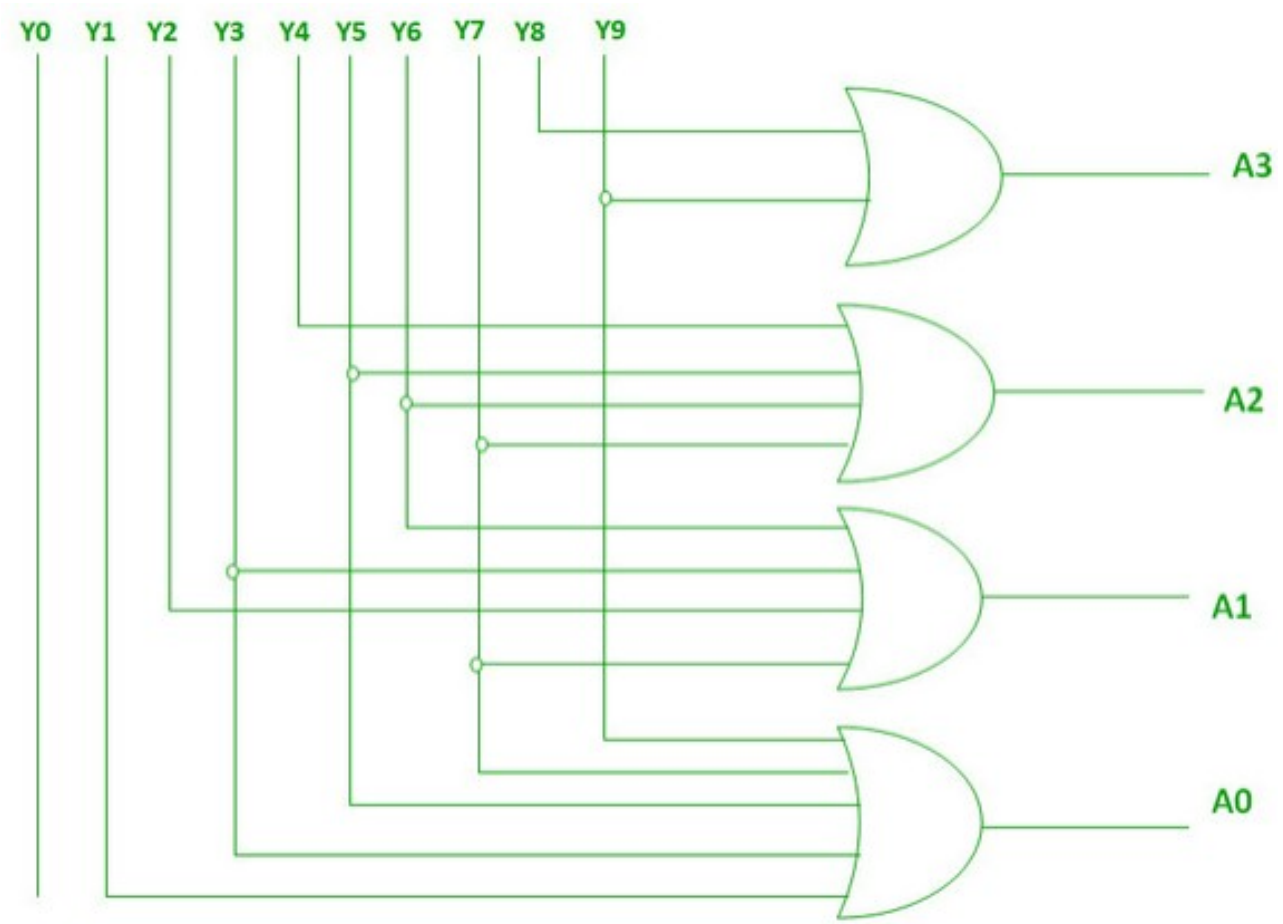
https://www.geeksforgeeks.org/encoder-in-digital-logic/

Codificador decimal para BCD

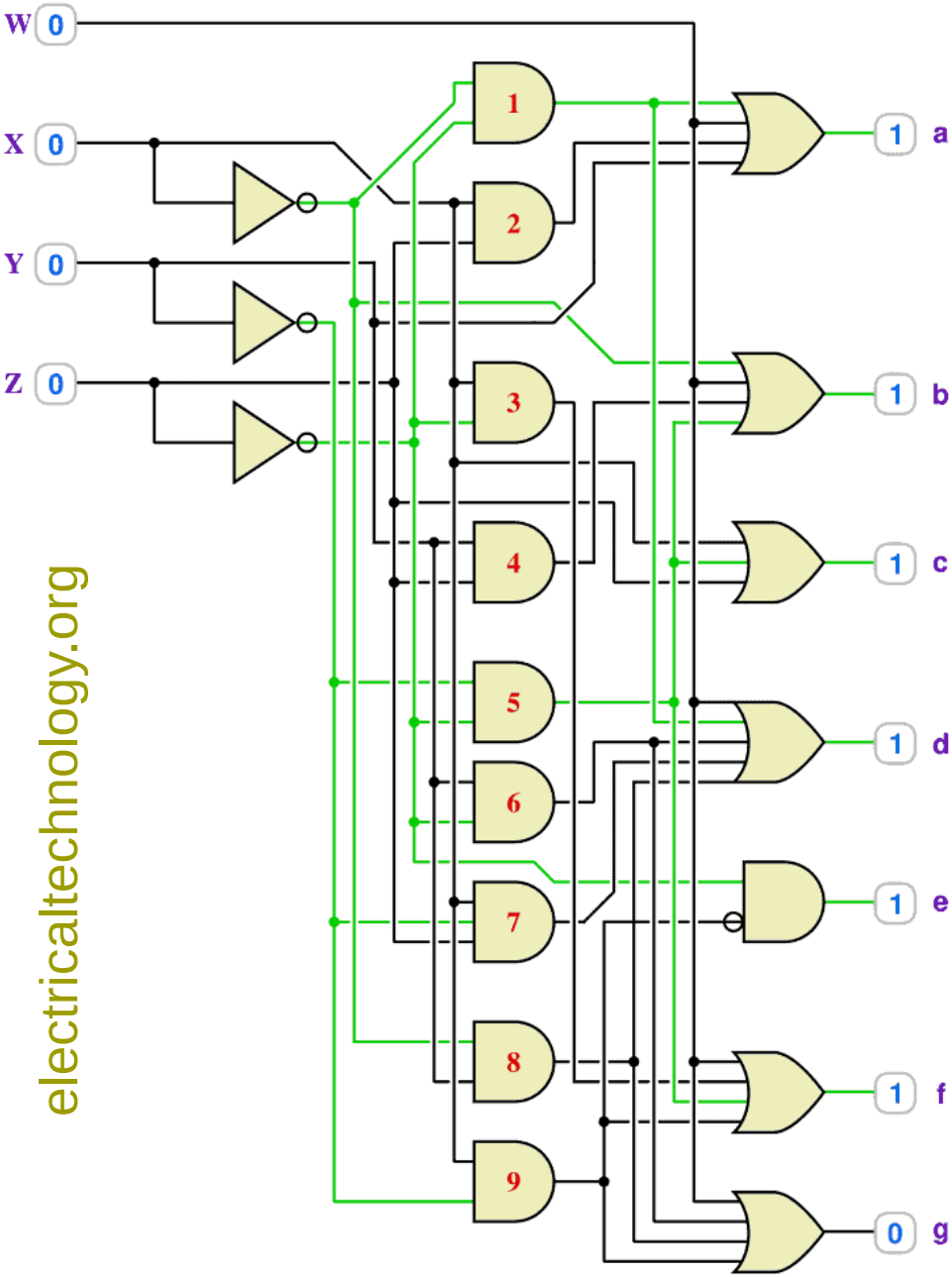
INPUTS										OUTPUTS			
Y9	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	A3	A2	A1	A0
0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	1

Codificador decimal para BCD

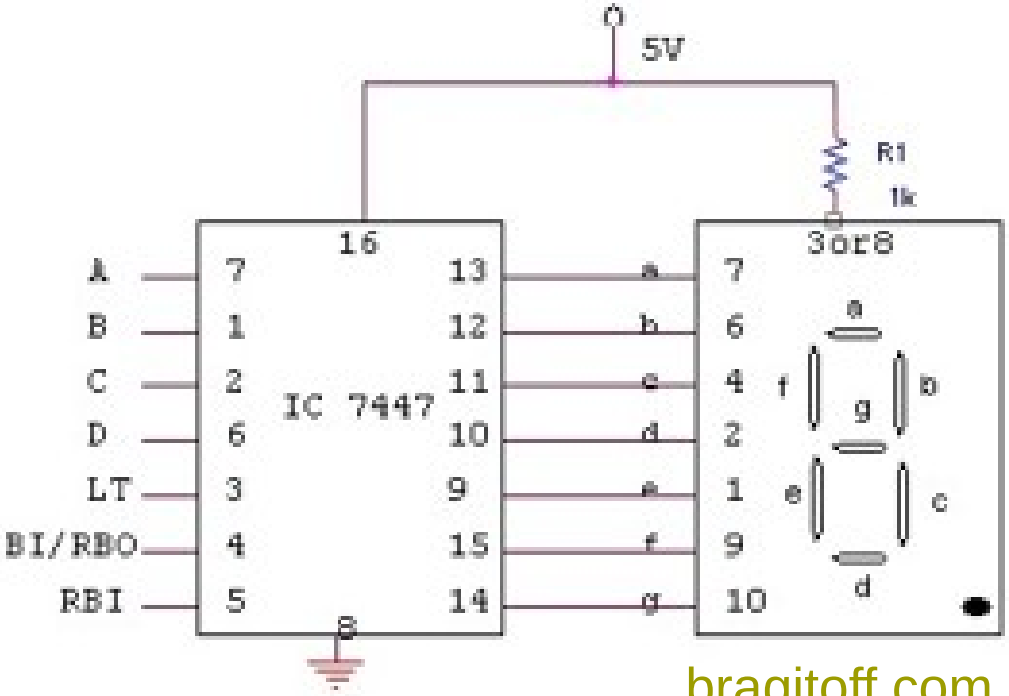
```
A3 = Y9 + Y8
A2 = Y7 + Y6 + Y5 +Y4
A1 = Y7 + Y6 + Y3 +Y2
A0 = Y9 + Y7 +Y5 +Y3 + Y1
```



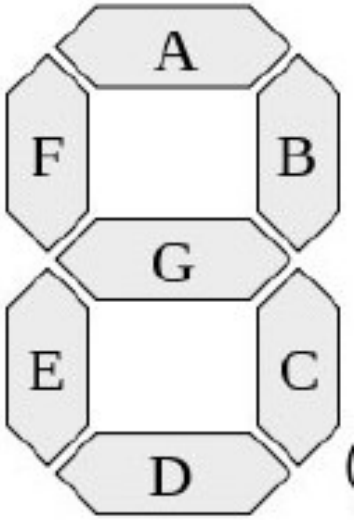
Decodificador BCD / 7 segmentos

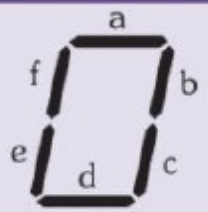
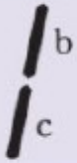
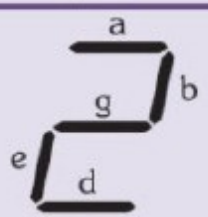
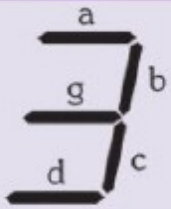
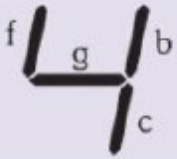
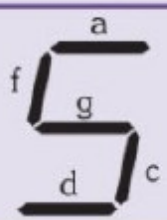


electricaltechnology.org



bragitoff.com



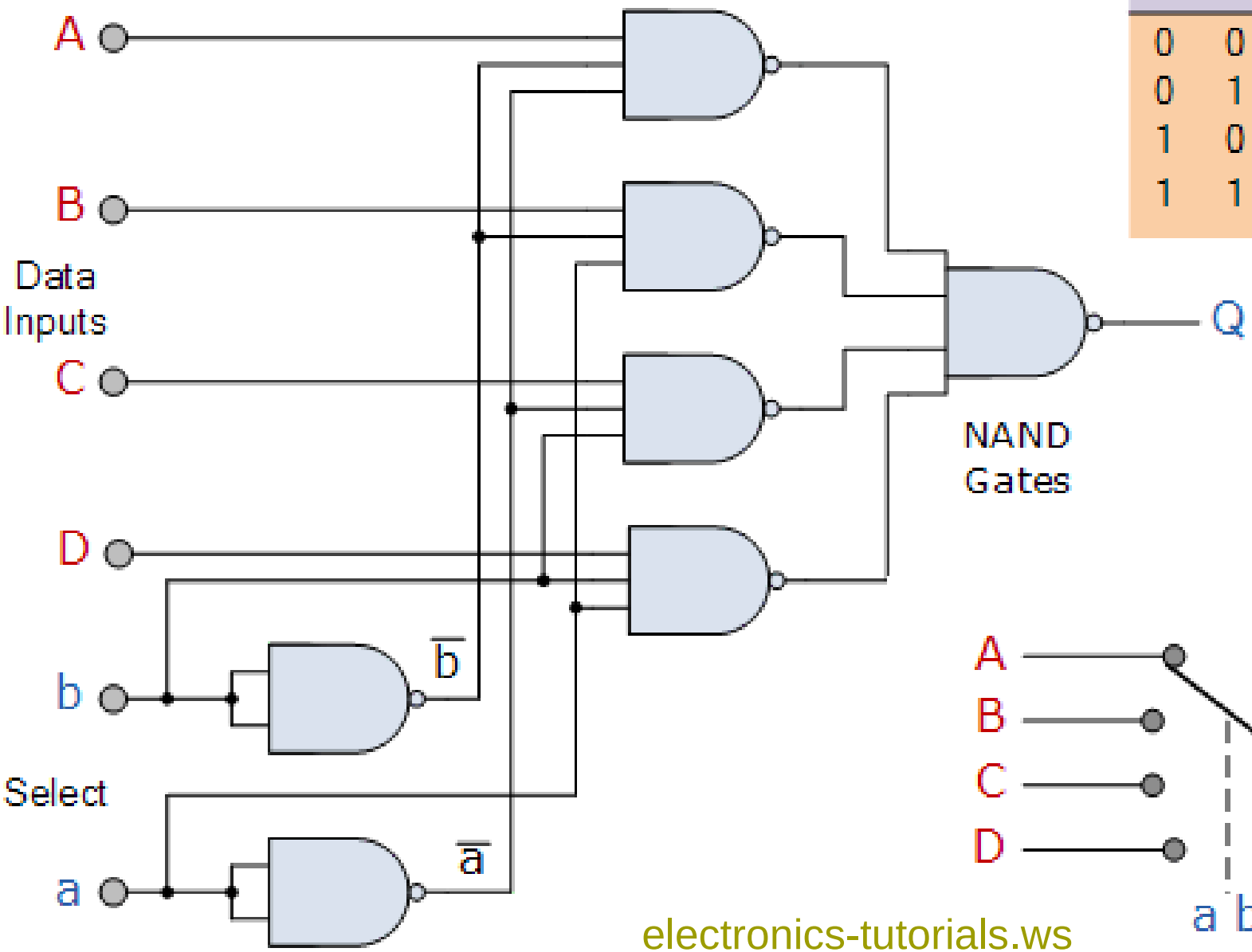
Caracteres	*Display*	BCD 8421				Código para 7 segmentos						
		A	B	C	D	a	b	c	d	e	f	g
0		0	0	0	0	1	1	1	1	1	1	0
1		0	0	0	1	0	1	1	0	0	0	0
2		0	0	1	0	1	1	0	1	1	0	1
3		0	0	1	1	1	1	1	1	0	0	1
4		0	1	0	0	0	1	1	0	0	1	1
5		0	1	0	1	1	0	1	1	0	1	1

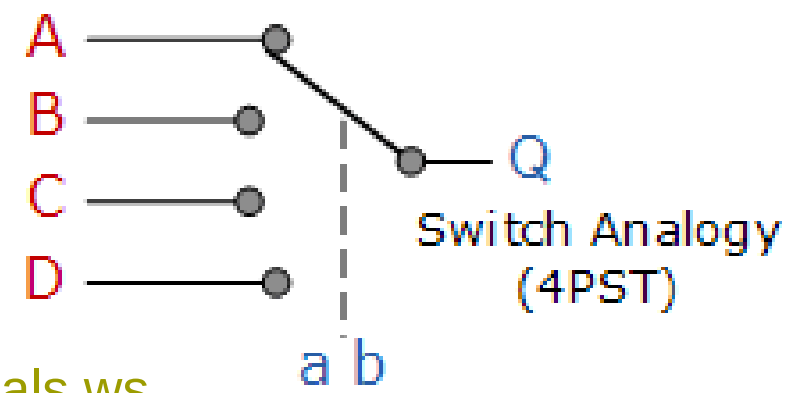
(e assim por diante)

Multiplexador (MUX)

Truth Table

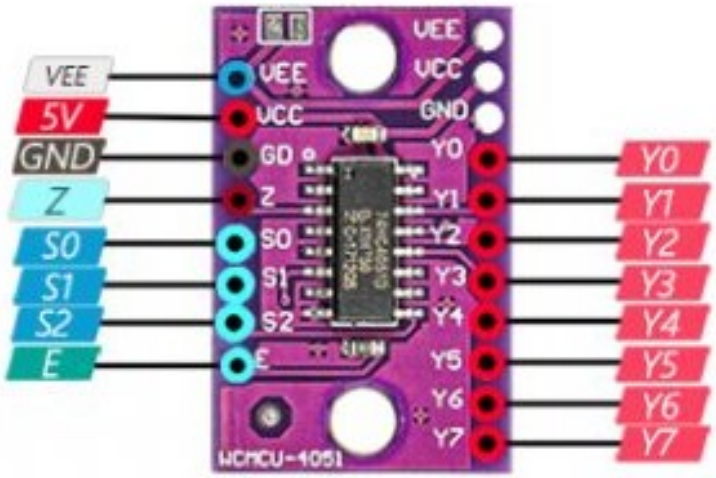
Select		Inputs				
b	a	D	C	B	A	Q
0	0	x	x	x	1	1
0	1	x	x	1	x	1
1	0	x	1	x	x	1
1	1	1	x	x	x	1

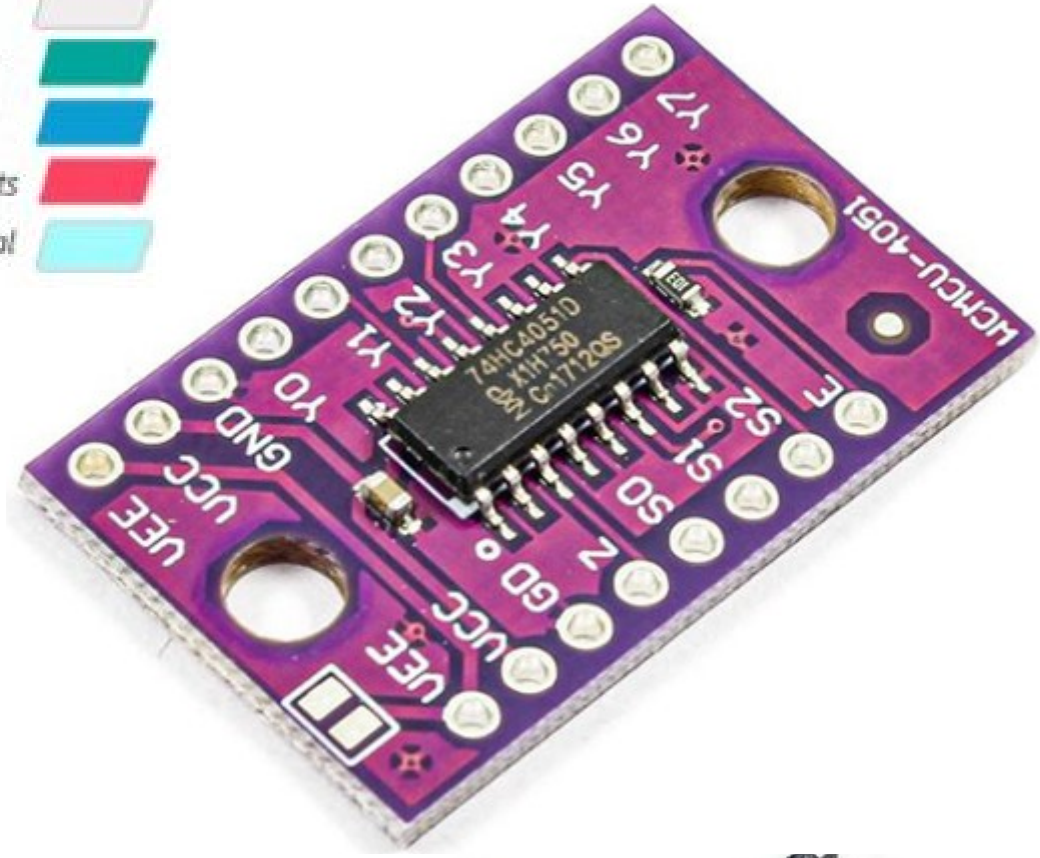


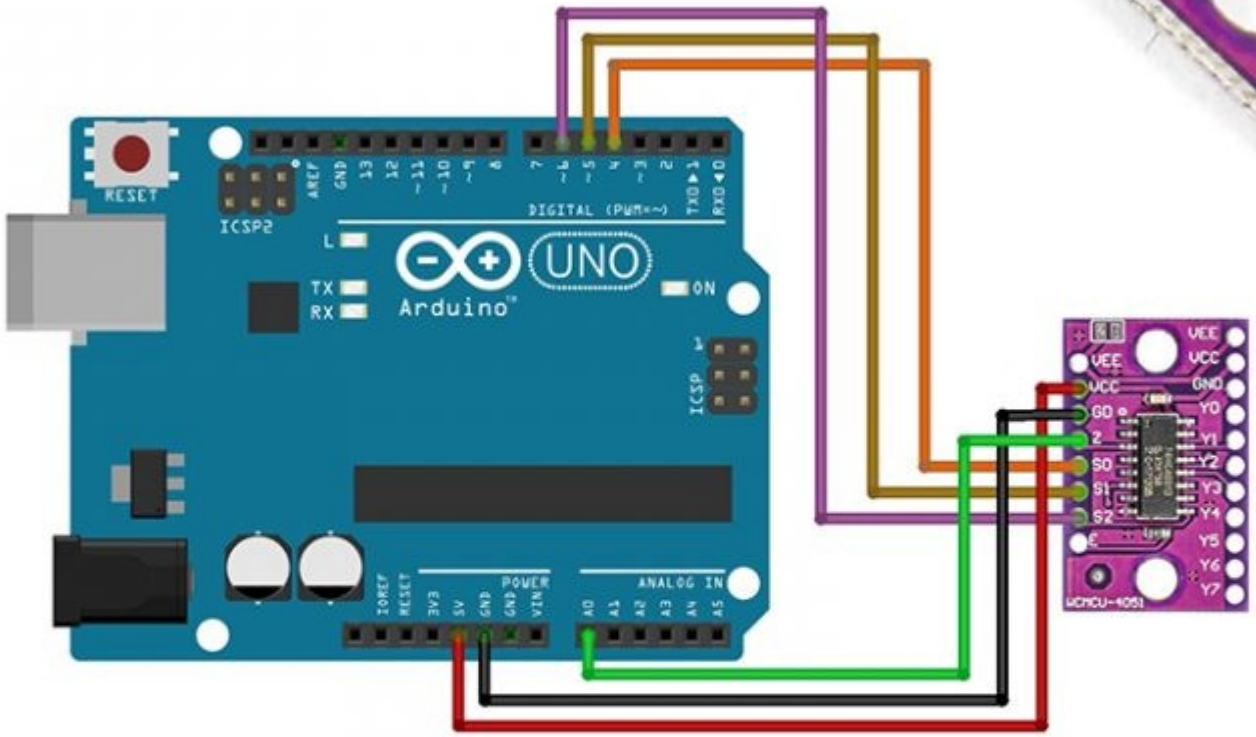


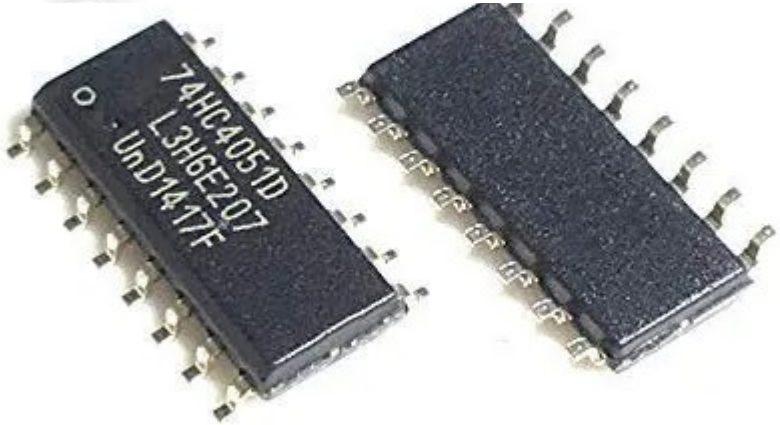
V_{EE} = tensão de referência comum, geralmente conectada ao nível de terra (0V)

74HC405 (de)multiplexador analógico de 8 canais



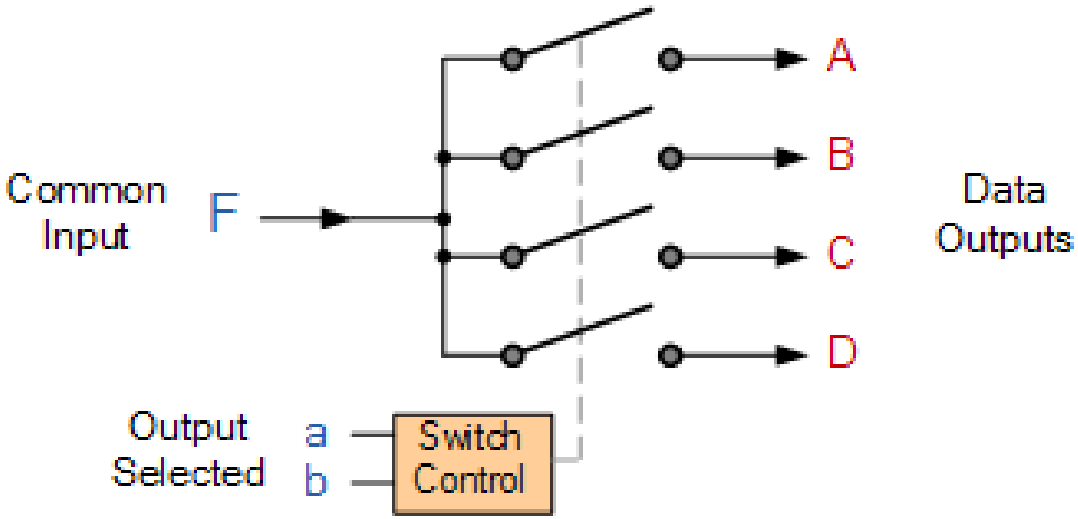




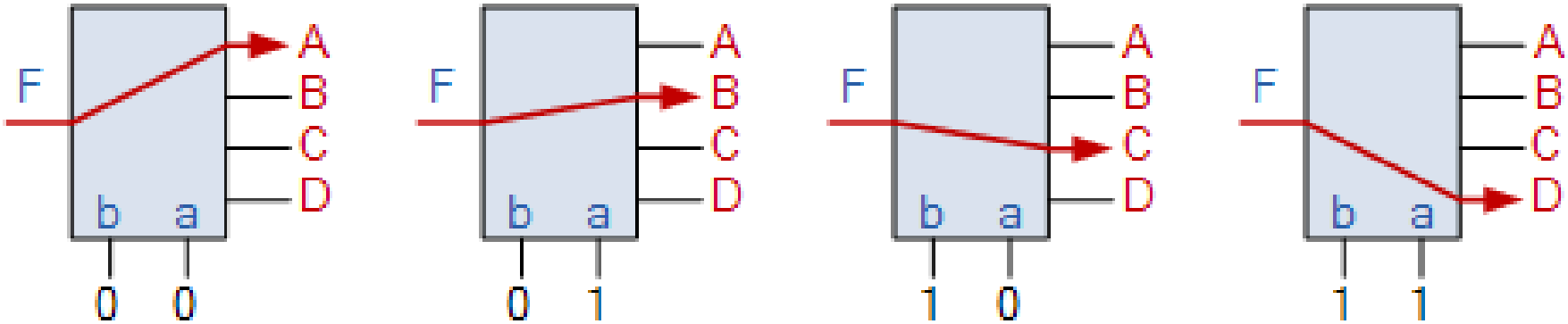


https://electropeak.com

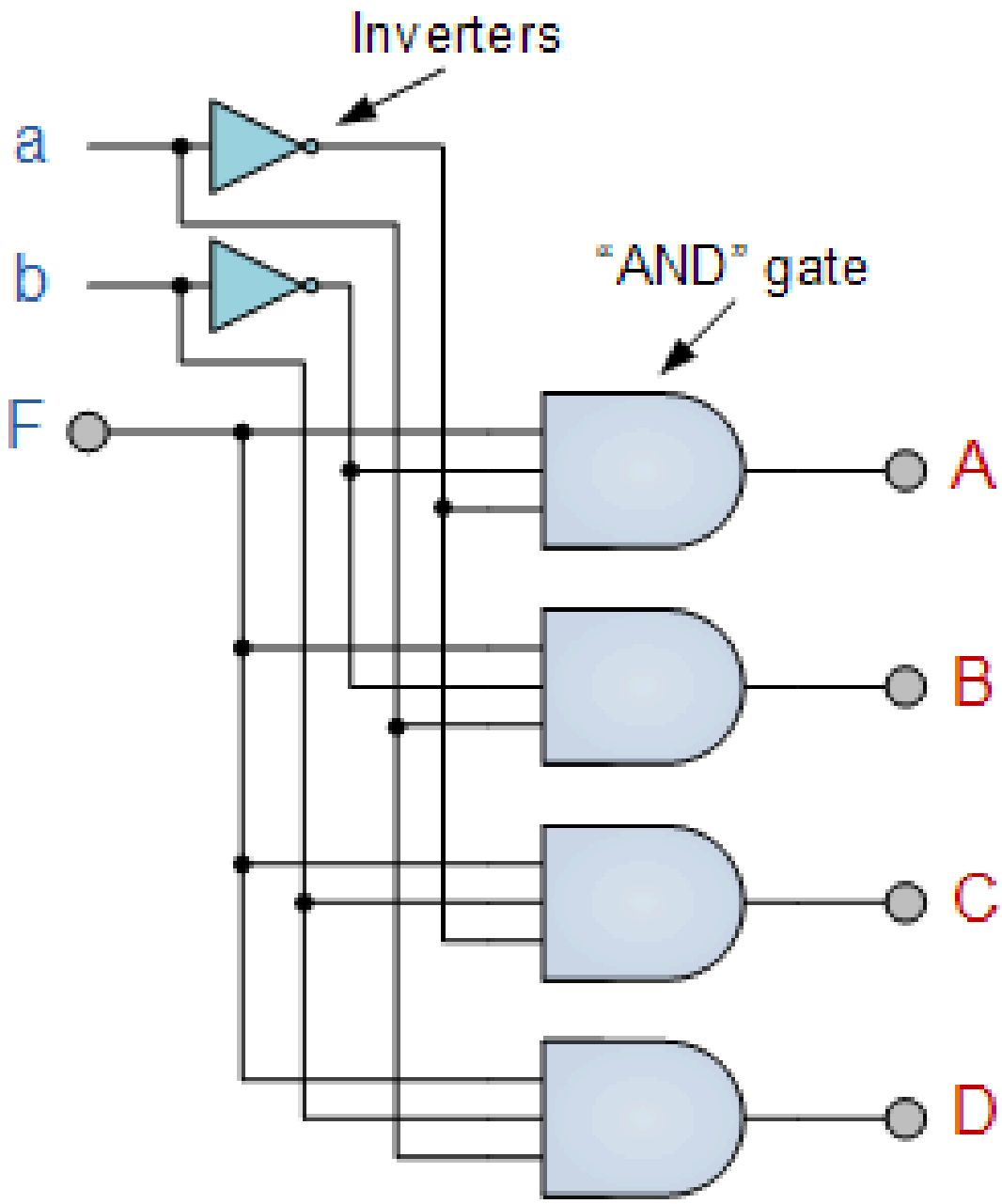
Demultiplexador (DEMUX)



Output Select		Data Output Selected
b	a	
0	0	A
0	1	B
1	0	C
1	1	D



DEMUX



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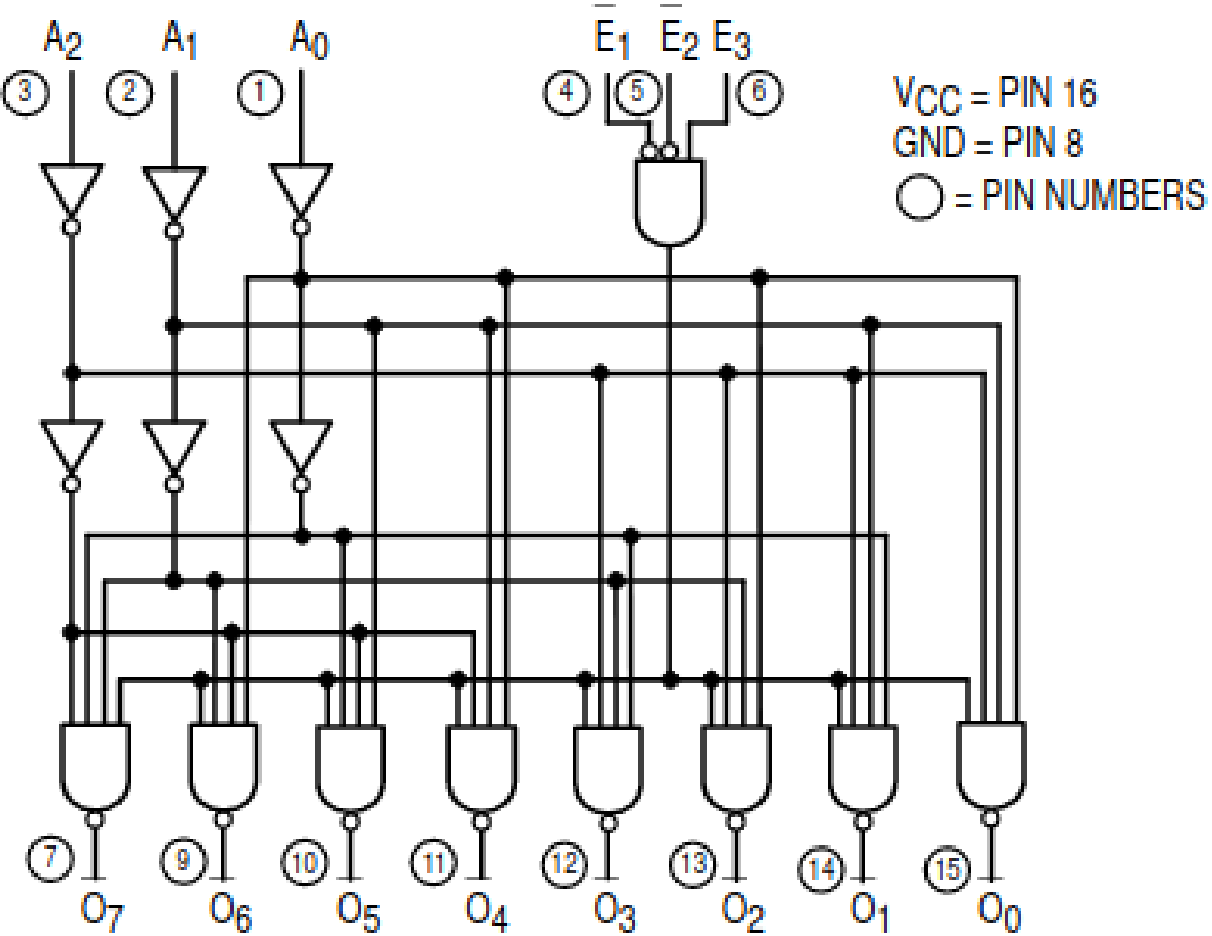
74LS138 1-OF-8 DECODER/ DEMULTIPLEXER



INPUTS						OUTPUTS							
E_1	E_2	E_3	A_0	A_1	A_2	O_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	H	L	L	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	H	H	L	H	H	H	L	H	H	H	H
L	L	H	L	L	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	L	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L